

# MARTINGALES

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## DON'T EXPECT A BUZZ, RETURNS ARE A LIGHTYEAR AWAY

In many stochastic models, we are interested not only in the possible future states of a system but also in its expected behaviour over time. A particularly important class of processes are martingales, which can be viewed as mathematical models of fair games. Informally, a martingale is a stochastic process whose future expected value, given all currently available information, is equal to its present value.

## RANDOM WALKS

A **random walk** is a stochastic process of the form,  $\{S_n\}_{n \geq 0}$ , where  $S_n = X_1 + X_2 + \dots + X_n$ , and  $\{X_i\}_{i \geq 1}$  are a sequence of IID random variables. The random variables,  $X_i$ , represent the successive increments of the process, while  $S_n$  records the cumulative total after  $n$  steps. The term comes from viewing  $S_n$  as the position of a particle or walker at time  $n$ , where at each step, the walker moves by a random amount determined by successive steps,  $X_1, X_2, \dots$ . If  $X_i$  takes values  $+1$  and  $-1$  with probabilities  $p$  and  $1 - p$  respectively,  $S_n$  is called a **simple random walk** with parameter  $p$ . Now, if  $p = \frac{1}{2}$ , i.e., upward and downward moves are equally likely, the process,  $S_n$ , is a **symmetric random walk**. For a symmetric random walk,  $\mathbb{E}[S_n] = 0$  and  $\text{Var}(S_n) = \mathbb{E}[S_n^2] - \mathbb{E}[S_n]^2 = \mathbb{E}[S_n^2] = n$ .

## MARTINGALES

Let us define what it is for a stochastic process to be a martingale. A stochastic process,  $\{X_n\}_{n \geq 0}$ , is called a **martingale** with respect to itself, that is, the filtration  $\{\mathcal{F}_n\}_{n \geq 0}$ , is:

- (i)  $\mathbb{E}[|X_n|] < \infty \forall n \geq 0$ ,
- (ii)  $\mathbb{E}[X_n | \mathcal{F}_{n-1}] = X_{n-1}$  for any  $n \geq 1$ .

Intuitively, a martingale is a stochastic process that does not change on average. We have the following two variations of a martingale:

- (i)' if  $\mathbb{E}[X_n | \mathcal{F}_{n-1}] \leq X_{n-1}$ , then it is called a **supermartingale**,
- (ii)' if  $\mathbb{E}[X_n | \mathcal{F}_{n-1}] \geq X_{n-1}$ , then it is called a **submartingale**.

If we have two stochastic processes,  $\{X_n\}_{n \geq 0}$  and  $\{Y_n\}_{n \geq 0}$ , defined on the same probability space, we call  $\{X_n\}_{n \geq 0}$  a martingale w.r.t.  $\{Y_n\}_{n \geq 0}$  if it satisfies (i) and (ii), with  $\{\mathcal{F}_n\}_{n \geq 0}$  being the filtration associated with  $\{Y_n\}_{n \geq 0}$ .

Do note that a martingale process need not be a Markov process; similarly, a Markov process need not be a martingale process.

## AN EXAMPLE: THE UNBIASED RANDOM WALK

Let  $X_n = \xi_1 + \xi_2 + \dots + \xi_n$ , where  $\{\xi_i\}_{i \geq 1}$  is a family of IID random variables, with  $\mathbb{P}(\xi_i = +1) = \mathbb{P}(\xi_i = -1) = \frac{1}{2}$ , for any  $i \geq 1$ , i.e., a symmetric random walk. Prove that the symmetric random walk is a martingale with respect to its natural filtration.

**Solution:** Note,  $\mathbb{E}[\xi] = (+1) \cdot \left(\frac{1}{2}\right) + (-1) \cdot \left(\frac{1}{2}\right) = 0$ . Set  $X_0 = 0$  and take  $\mathcal{F}_0 = \{\emptyset, \Omega\}$ . To verify that  $\{X_n\}_{n \geq 0}$  is a martingale w.r.t. itself, we first verify (i). For  $n > 0$  we have  $\mathbb{E}[|X_n|] = \mathbb{E}[|\xi_1 + \xi_2 + \dots + \xi_n|] \leq \mathbb{E}[|\xi_1| + |\xi_2| + \dots + |\xi_n|] = \mathbb{E}[n] = n < \infty$ , whereas  $\mathbb{E}[|X_0|] < \infty$  too. We can now verify (ii). For  $n > 0$  we have:

$$\begin{aligned}
 \mathbb{E}[X_n | \mathcal{F}_{n-1}] &= \mathbb{E}[X_{n-1} + \xi_n | \mathcal{F}_{n-1}] \\
 &= \mathbb{E}[X_{n-1} | \mathcal{F}_{n-1}] + \mathbb{E}[\xi_n | \mathcal{F}_{n-1}], \text{ by linearity of the conditional expectation} \\
 &= X_{n-1} + \mathbb{E}[\xi_n | \mathcal{F}_{n-1}], \text{ since } X_{n-1} \text{ is measurable in } \mathcal{F}_{n-1} \\
 &= X_{n-1} + \mathbb{E}[\xi_n], \text{ since } \xi_n \text{ is independent of } X_0, X_1, \dots, X_{n-1} \\
 &= X_{n-1}, \text{ since } \mathbb{E}[\xi_n] = 0.
 \end{aligned}$$

You could argue that this is the most intuitive martingale example.

## CONCLUSION

Martingales are stochastic processes where the conditional expected future value is equal to their current value. Although martingales and Markov processes are distinct concepts, both are key in the study of stochastic systems.

The symmetric random walk is possibly one of the simplest and most intuitive examples of a martingale. Other examples could include the product martingale, the discrete stochastic integral, *etc.*

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